



SAFETY DATA SHEET

QSR MOISTURIZING HAND SANITIZER

1. PRODUCT AND COMPANY IDENTIFICATION

Product name : 큐에스알 모이스처라이징 핸드 새니타이저
QSR MOISTURIZING HAND SANITIZER

Other means of identification : Not applicable

Recommended use : Hand Sanitizer

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : Product is sold ready to use.

Company : ECOLAB KOREA LTD.
8F, Daemyung Tower
135 Beobwon-ro, Songpa-gu, Seoul Korea 05836
82-1588-7595

Emergency telephone : 82-1588-7595

Issuing date : 31.01.2019

2. HAZARDS IDENTIFICATION

GHS Classification

Flammable liquids : Category 3
Serious eye damage/eye irritation : Category 2

GHS label elements

Hazard pictograms :



Signal Word : Warning

Hazard Statements : Flammable liquid and vapor.
Causes serious eye irritation.

Precautionary Statements : **Prevention:**
Keep away from heat/sparks/open flames/hot surfaces. No smoking.
Keep container tightly closed.
Response:
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/ attention. In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish.
Storage:
Store in a well-ventilated place. Keep cool.
Disposal:
Dispose of contents and container according to wastes control act.

Other hazards : None known.

3. COMPOSITION/INFORMATION ON INGREDIENTS

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Pure substance/mixture : Mixture

Chemical name	CAS-No.	Concentration (%)
ethanol	64-17-5	60 - 70
glycerin	56-81-5	1 - 10
2-propanol, 1,1',1'',1'''-(1,2-ethanediyldinitrilo)tetrakis-2-methylpropan-2-ol	102-60-3	1 - 10
	75-65-0	0.1 - 1

All unlisted components do not contribute to the overall hazard of the product and are trade secret, making up 100% of the full composition.

4. FIRST AID MEASURES

In case of eye contact : Rinse with water.

In case of skin contact : Rinse with water.

If swallowed : Rinse mouth. Get medical attention if symptoms occur.

If inhaled : Get medical attention if symptoms occur.

Protection of first-aiders : No special precautions are necessary for first aid responders.

Notes to physician : Treat symptomatically.

Most important symptoms and effects, both acute and delayed : See Section 11 for more detailed information on health effects and symptoms.

5. FIRE-FIGHTING MEASURES

Suitable extinguishing media : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable extinguishing media : High volume water jet

Specific hazards during fire fighting : Fire Hazard
Keep away from heat and sources of ignition.
Flash back possible over considerable distance.
Beware of vapors accumulating to form explosive concentrations.
Vapors can accumulate in low areas.

Hazardous combustion products : Decomposition products may include the following materials:
Carbon oxides

Special protective equipment for fire-fighters : Use personal protective equipment.

Specific extinguishing methods : Use water spray to cool unopened containers. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

6. ACCIDENTAL RELEASE MEASURES

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- Personal precautions, protective equipment and emergency procedures : Remove all sources of ignition. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.
- Environmental precautions : Do not allow contact with soil, surface or ground water.
- Methods and materials for containment and cleaning up : Eliminate all ignition sources if safe to do so. Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

7. HANDLING AND STORAGE

- Advice on safe handling : Keep away from fire, sparks and heated surfaces. Take necessary action to avoid static electricity discharge (which might cause ignition of organic vapors). Open drum carefully as content may be under pressure.
- Conditions for safe storage : Keep away from heat and sources of ignition. Keep in a cool, well-ventilated place. Keep away from oxidizing agents. Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.
- Storage temperature : 0 °C to 35 °C

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Components	CAS-No.	Form of exposure	Permissible concentration	Basis
ethanol	64-17-5	TWA	1,000 ppm	KR OEL
ethanol	64-17-5	TWA	1,000 ppm 1,900 mg/m ³	NIOSH REL
		TWA	1,000 ppm 1,900 mg/m ³	OSHA Z1
glycerin	56-81-5	TWA (Mist)	10 mg/m ³	KR OEL
glycerin	56-81-5	TWA	10 mg/m ³	ACGIH
		TWA (respirable fraction)	5 mg/m ³	OSHA Z1
2-methylpropan-2-ol	75-65-0	STEL	150 ppm	KR OEL
		TWA	100 ppm	KR OEL
2-methylpropan-2-ol	75-65-0	TWA	100 ppm	ACGIH
		TWA	100 ppm 300 mg/m ³	NIOSH REL
		STEL	150 ppm 450 mg/m ³	NIOSH REL
		TWA	100 ppm 300 mg/m ³	OSHA Z1

- Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

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Personal protective equipment

Eye protection	: No special protective equipment required.
Hand protection	: No special protective equipment required.
Skin protection	: No special protective equipment required.
Respiratory protection	: No personal respiratory protective equipment normally required.
Hygiene measures	: No specific measures identified.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	: gel
Color	: cloudy, colorless
Odor	: alcohol-like
pH	: 6.0 - 8.5, (100 %)
Flash point	: 24 °C closed cup
Odor Threshold	: No data available
Melting point/freezing point	: No data available
Initial boiling point and boiling range	: < 35 °C
Evaporation rate	: No data available
Flammability (solid, gas)	: No data available
Upper explosion limit	: No data available
Lower explosion limit	: No data available
Vapor pressure	: No data available
Relative vapor density	: No data available
Relative density	: 0.8 - 0.94
Water solubility	: soluble
Solubility in other solvents	: No data available
Partition coefficient: n-octanol/water	: No data available
Autoignition temperature	: No data available
Thermal decomposition	: No data available
Viscosity, kinematic	: No data available
Explosive properties	: No data available
Oxidizing properties	: The substance or mixture is not classified as oxidizing.
Molecular weight	: No data available
VOC	: No data available

10. STABILITY AND REACTIVITY

Chemical stability	: Stable under normal conditions.
Possibility of hazardous	: No dangerous reaction known under conditions of normal use.

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reactions

Conditions to avoid : Heat, flames and sparks.

Incompatible materials : None known.

Hazardous decomposition products : In case of fire hazardous decomposition products may be produced such as:
Carbon oxides

11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Potential Health Effects

Eyes : Causes serious eye irritation.

Skin : Health injuries are not known or expected under normal use.

Ingestion : Health injuries are not known or expected under normal use.

Inhalation : Health injuries are not known or expected under normal use.

Chronic Exposure : Health injuries are not known or expected under normal use.

Experience with human exposure

Eye contact : Redness, Pain, Irritation

Skin contact : No symptoms known or expected.

Ingestion : No symptoms known or expected.

Inhalation : No symptoms known or expected.

Toxicity

Product

Acute oral toxicity : No data available

Acute inhalation toxicity : No data available

Acute dermal toxicity : No data available

Skin corrosion/irritation : No data available

Serious eye damage/eye irritation : No data available

Respiratory or skin sensitization : No data available

Carcinogenicity : No data available

Reproductive effects : No data available

Germ cell mutagenicity : No data available

Teratogenicity : No data available

STOT-single exposure : No data available

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STOT-repeated exposure : No data available

Aspiration toxicity : No data available

Components

Acute oral toxicity : ethanol
LD50 Rat: 10,470 mg/kg

glycerin
LD50 Rat: 18,300 mg/kg

2-propanol, 1,1',1'',1'''-(1,2-ethanediyl)dinitrilo)tetrakis-
LD50 Rat: 2,890 mg/kg

2-methylpropan-2-ol
LD50 Rat: 2,733 mg/kg

Components

Acute inhalation toxicity : ethanol
4 h LC50 Rat: 117 mg/l Test atmosphere: vapor

Components

Acute dermal toxicity : ethanol
LD50 Rabbit: > 15,800 mg/kg

glycerin
LD50 Rabbit: 23,000 mg/kg

12. ECOLOGICAL INFORMATION

Ecotoxicity

Environmental Effects : This product has no known ecotoxicological effects.

Product

Toxicity to fish : No data available

Toxicity to daphnia and other aquatic invertebrates : No data available

Toxicity to algae : No data available

Components

Toxicity to fish : ethanol
96 h LC50 Pimephales promelas: > 100 mg/l

glycerin
96 h LC50 Fish: 855 mg/l

2-propanol, 1,1',1'',1'''-(1,2-ethanediyl)dinitrilo)tetrakis-
96 h LC50: 2,200 mg/l

Components

Toxicity to daphnia and other aquatic invertebrates : 2-methylpropan-2-ol
48 h EC50: 933 mg/l

Persistence and degradability

Biodegradable

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Bioaccumulative potential

No data available

Mobility in soil

No data available

Other adverse effects

No data available

13. DISPOSAL CONSIDERATIONS

- Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.
- Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

14. TRANSPORT INFORMATION

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport (KR_DG)

UN number : 1170
Description of the goods : ETHANOL SOLUTION
Class : 3
Packing group : III
Environmentally hazardous : no

Sea transport (IMDG/IMO)

UN number : 1170
Description of the goods : ETHANOL SOLUTION
Class : 3
Packing group : III
Marine pollutant : no

Special precautions for user : None

15. REGULATORY INFORMATION

National regulatory information

Regulation under the Occupational Safety and Health Act

Harmful Substances Prohibited from Manufacturing

Not applicable

Harmful Substances Required Permission for Manufacture

Not applicable

Hazardous substances requiring management

Not applicable

Controlled Substances Subject to Environment Monitoring

Not applicable

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Controlled Substances Subject to Health Examination

Not applicable

Substances with Workplace Exposure Limits

ethanol

glycerin

2-methylpropan-2-ol

Act on the Registration and Evaluation, etc. of Chemical Substances, Chemicals Control Act

Toxic Chemicals

Not applicable

Restricted Chemicals

Not applicable

Prohibited Chemicals

Not applicable

Toxic Release Inventory

Not applicable

Accident Precaution Chemicals

Not applicable

Dangerous Substances Safety Management Act

Classification : Group 4, Flammable liquids, Alcohols

Hazard rank : Hazardous rank II

Designated Quantity : 400 liter

Warning : Keep away from fire

Waste Control Act

Industrial waste

The ingredients of this product are reported in the following inventories:

United States TSCA Inventory :

not determined

Canadian Domestic Substances List (DSL) :

All components of this product are on the Canadian DSL

Australia Inventory of Chemical Substances (AICS) :

not determined

New Zealand. Inventory of Chemical Substances :

not determined

Japan. ENCS - Existing and New Chemical Substances Inventory :

not determined

Korea. Korean Existing Chemicals Inventory (KECI) :

On the inventory, or in compliance with the inventory

Philippines Inventory of Chemicals and Chemical Substances (PICCS) :

not determined

China. Inventory of Existing Chemical Substances in China (IECSC) :

not determined

Taiwan Chemical Substance Inventory (TCSI) :

not determined

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16. OTHER INFORMATION

Sources of key data used to compile the Safety Data Sheet
Globally Harmonized System of Classification and Labeling of Chemicals (GHS)
IARC: (International Agency for Research on Cancer)
US. National Toxicology Program (NTP) Report on Carcinogens
ECHA List of Publishable Substances Registered
EU HPVCs (High Production Volume Chemicals)

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Prepared by : Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Material Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.